

Exercises III : Topology III

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1. a) Suppose $E \rightarrow M$ is a real vector bundle of dimension n with a metric.
 - (i) There is a zero section $s : M \rightarrow E$ which takes each point x to the 0 in E_x . Show that E deformation retracts onto the zero section.
 - (ii) Prove that the induced sphere bundle $S(E) \rightarrow M$ is a fibre bundle with fibre S^{n-1} .
 - (iii) Prove that $E - s(M)/\mathbb{R}^\times = S(E)/(x \simeq -x)$ gives a fibre bundle $P(E) \rightarrow M$ with fibre $\mathbb{R}P^{n-1}$.
 - (iv) Prove that the frame bundle $F(E) = \{(x, v_1, \dots, v_n) \in M \times E^n \mid v_i \in E_x, \langle v_i, v_j \rangle = \delta_{ij}\}$ gives a fibre bundle $F(E) \rightarrow M$ with fibre $O(n)$.
 - b) Suppose $E \rightarrow M$ is a complex vector bundle of dimension n with a hermitian metric.
 - (i) Prove that the induced sphere bundle $S(E) \rightarrow M$ is a fibre bundle with fibre S^{2n-1} .
 - (ii) Prove that $E - s(M)/\mathbb{C}^\times = S(E)/S^1$ gives a fibre bundle $P(E) \rightarrow M$ with fibre $\mathbb{C}P^{n-1}$.
 - (iii) Prove that the frame bundle $F(E) = \{(x, v_1, \dots, v_n) \in M \times E^n \mid v_i \in E_x, \langle v_i, v_j \rangle = \delta_{ij}\}$ gives a fibre bundle $F(E) \rightarrow M$ with fibre $U(n)$.
 2. Suppose ξ has an inner product $\langle \cdot, \cdot \rangle$. Prove that if ξ is a real bundle, an inner product gives an isomorphism $\xi \cong \xi^* = \text{Hom}(\xi, \mathbb{R})$. In the complex case we have an isomorphism $\bar{\xi} \cong \xi^* = \text{Hom}(\xi, \mathbb{C})$.
 3. Write η for the canonical line bundle (complex) over $\mathbb{C}P^n$ and γ for the canonical real line bundle over $\mathbb{R}P^n$. Consider the natural quotient map from $\pi : \mathbb{R}P^{2n+1} \rightarrow \mathbb{C}P^n$ defined by $[x_0, x_1, \dots, x_{2n}, x_{2n+1}] \mapsto [x_0 + ix_1, \dots, x_{2n} + ix_{2n+1}]$. Show that $\pi^*(\eta) \cong c(\gamma)$ (c stands for the complexification functor $V \mapsto V \otimes \mathbb{C}$) and $r(\pi^*\eta) \cong \gamma \oplus \gamma$ (r is the functor which takes a complex vector bundle to the real bundle forgetting the complex structure).
 4. Let r be the functor which takes a complex vector bundle to its underlying real vector bundle. Let c be the functor which takes a real vector bundle ξ to $\xi \otimes_{\mathbb{R}} \mathbb{C}$.
 - a) Prove that $rc(\xi) = \xi \oplus \xi$.
 - b) Prove that $cr(\xi) = \xi \oplus \bar{\xi}$.
 5. Prove that every 2-dimensional vector bundle (over a paracompact base) is the underlying real bundle of a complex line bundle.
 6. Prove that any complex vector bundle over a 1-dimensional CW complex is trivial.
 7. Prove that every complex line bundle over S^2 is isomorphic to $\gamma^{\otimes k}$ or $(\gamma^*)^{\otimes k}$ for some $k \geq 0$. (Here, γ denotes the canonical line bundle over $\mathbb{C}P^1 \cong S^2$.)
 8. Prove that if a complex line bundle over a CW complex is trivial over the 2-skeleton, then it is trivial.
 9. Consider the $\mathbb{Z}/2$ -action on $\mathbb{C}P^n$ by conjugation. Let $X_n = S^\infty \times_{\mathbb{Z}/2} \mathbb{C}P^n$. Prove that X_n is the projectivisation of a complex vector bundle over $\mathbb{R}P^\infty$.